

Worksheet 14.3: Solutions

Corrosion of iron

No.	Answer										
1	a $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^{-}$ b $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4\text{e}^{-} \rightarrow 4\text{OH}^{-}(\text{aq})$										
2	The dessicant (anhydrous calcium chloride) removes water. With no water present, corrosion cannot occur.										
3	Boiling removes oxygen from the water. The oil prevents any further oxygen dissolution. With no oxygen present, corrosion cannot occur.										
4	Pure oxygen, rather than air, means a higher concentration of the reactant oxygen. Reaction rate increases with increasing reactant concentration.										
5	a No oxygen present. b There is very little dissolved oxygen in the deepest parts of the ocean. c The oil prevents contact between the oxygen and water in the air, and the iron surface. d The magnesium is more reactive (stronger reductant) than iron, so the magnesium, rather than the iron, will supply the electrons for the reduction of oxygen. e The negative terminal supplies electrons to the surface of the iron. With an excess of electrons present, oxidation of the iron is prevented.										
6	a The zinc acts as a mechanical barrier, preventing contact between the iron and the oxygen/water in the air. Zinc can also act as a sacrificial anode, being oxidised in preference to the iron. b The tin acts as a mechanical barrier, preventing contact between the iron and the oxygen/water in the air. c Zinc may react with foods and may be toxic.										
7	<table border="0"> <tr> <td>Gardening secateurs</td><td>Coating with oil</td></tr> <tr> <td>Natural gas pipeline</td><td>Attachment to sacrificial anodic magnesium blocks</td></tr> <tr> <td>Wharf</td><td>Cathodic protection with a DC power supply</td></tr> <tr> <td>Patio furniture</td><td>Painting</td></tr> <tr> <td>A shed roof</td><td>Galvanising</td></tr> </table>	Gardening secateurs	Coating with oil	Natural gas pipeline	Attachment to sacrificial anodic magnesium blocks	Wharf	Cathodic protection with a DC power supply	Patio furniture	Painting	A shed roof	Galvanising
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	DRY CELL		LEAD ACID ACCUMULATOR	FUEL CELL
ANODE	Zinc		Lead	Carbon
CATHODE	Graphite		Lead	Carbon
ELECTROLYTE	Zinc chloride/ammonium chloride		Sulfuric acid	HCl (acidic) or KOH (alkaline)
EQUATIONS: OXIDATION	$\text{Zn}_{(s)} \rightarrow \text{Zn}^{2+}_{(aq)} + 2e^{-}$		$\text{Pb}_{(s)} + \text{SO}_4^{2-}_{(aq)} \rightarrow \text{PbSO}_{4(s)} + 2e^{-}$	$\text{H}_{2(g)} \rightarrow 2\text{H}^{+}_{(aq)} + 2e^{-}$ or $\text{H}_{2(g)} + 2\text{OH}^{-}_{(aq)} \rightarrow 2\text{H}_2\text{O}_{(l)} + 2e^{-}$
REDUCTION	$2\text{MnO}_{2(s)} + 2\text{NH}_4^{+}_{(aq)} + 2e^{-} \rightarrow \text{Mn}_2\text{O}_{3(s)} + 2\text{NH}_{3(aq)} + \text{H}_2\text{O}_{(l)}$		$\text{PbO}_{2(s)} + \text{SO}_4^{2-}_{(aq)} + 4\text{H}^{+}_{(aq)} + 2e^{-} \rightarrow \text{PbSO}_{4(s)} + 2\text{H}_2\text{O}_{(l)}$	$\text{O}_{2(g)} + 4\text{H}^{+}_{(aq)} + 4e^{-} \rightarrow 2\text{H}_2\text{O}_{(l)}$ or $\text{O}_{2(g)} + 2\text{H}_2\text{O}_{(l)} + 4e^{-} \rightarrow 4\text{OH}^{-}_{(aq)}$
OVERALL	$\text{Zn}_{(s)} + 2\text{MnO}_{2(s)} + 2\text{NH}_4^{+}_{(aq)} \rightarrow \text{Zn}^{2+}_{(aq)} + \text{Mn}_2\text{O}_{3(s)} + 2\text{NH}_{3(aq)} + \text{H}_2\text{O}_{(l)}$		$\text{Pb}_{(s)} + \text{PbO}_{2(s)} + 2\text{SO}_4^{2-}_{(aq)} + 4\text{H}^{+}_{(aq)} \rightarrow 2\text{PbSO}_{4(s)} + 2\text{H}_2\text{O}_{(l)}$	$2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(l)}$
USES/OTHER	Primary cell-not rechargeable Limited shelf life Voltage of 1.5V Alkaline version available		Secondary cell-rechargeable; reverse of overall eqn., consumes water. Electrolyte density drops during discharge. Comprise 6 x 2V cells = 12V	Reactants fed in (not stored in cell) and converted to energy. Expensive. Used in spacecraft, produces water for drinking.